helmets, safety shoes, rubber gloves, safety belt etc., whichever is required during execution of work. The work is to be carried out at height, adequate safety is to be taken for the same.

33. Railways will no way be responsible for any loss of life or any injury caused to any of the Contractor's Staffs while executing the works. The liability for any compensation on account of injury sustained by an employee of the agency will be exclusively that of the Contractor.
34. The work shall be carried by the contractor confirming to latest ISI & IE rules &Workmanship should be of appropriate quality
35. Any damage caused by the firm to the machine structure, railway property or samples, necessary amount will be deducted from the bill as directed by competent authority.
36. All power and control wires/cables are to be replaced with new reputed make and acceptable size of conductor, insulation based on electrical loads.
37. Proper lugging, crimping, marking of control and power cable is required to be done by the contractor.
38. All power and control cables, lighting and other cables are to be tagged /ferruled at both ends with cable number indicated clearly.
39. The testing of wiring and cabling shall be made as per the electricity rule.
40. The Firm must provide all the detailed drawings, specification sheets, test certificates, circuit drawings of electrical components, warranty certificates etc.
41. Training for minimum 02 Nos. Supervisor (JE/SSE) and 04 technician staff shall be arranged at OEM or reputed training center.
42. Supply and providing rubber mat (as per voltage system up to 33KV) in front of VCB panels, The material shall be as per relevant IS.
43. Necessary supply of control cables/wires for control supply connections as well protection and indication as per site requirement shall be provided by contractor. Necessary control wiring connection modifications shall be carried as per site requirement.
44. Contactor shall provide first aid box, resuscitation chart and tool box with tools (as per requirement) in each sub-station.
45. All the release material shall be handed over to Railway if any.

(II) TECHNICAL SPECIFICATION

For Schedule-A

1. **For Item No. 1& 2**

   Supply, erection, testing and commissioning of 22 KV / 415 V 2 nos. 250 KVA Compact substation comprising of 2nos. Incoming and 2 nos. outgoing HT switchgear and LT switchgears including foundation.

   Supply, loading, transportation and unloading to site, erection, testing and commissioning of combined Packaged / Compact Substation comprising of 2 nos. 250 KVA, 22 KV / 433 V transformers, inbuilt four way Compact type SF6 RMU HT switchgear and LT switchgear panel. The transformer shall have dry type cast resin, copper wound outdoor type distribution Transformer with off circuit tapings, ANAN cooling & vector diagram DYN 11 with all standard equipment's and fittings such as Digital Winding temperature scanner with alarm and tripping settings. The price shall also cover cost of OEM recommended plinth (base) for PSS. The PSS including transformer, HT & LT Switchgears shall be designed, manufactured and tested (all type) in accordance with the latest relevant IS & IEC (viz. IS 11171, IEC 600726, IEC 62271-200, IEC 62271-202, IEC 61439 1-6, IEC 60947-2 or latest) standards and specification enclosed. The contractor shall submit original test report from CPRI / ERDA / Any other equivalent Govt. Testing Agencies. The complete Compact / Packaged Substation should be got approved by Dy.CEE(W)/PL before supply.

TECHNICAL SPECIFICATION FOR 250 KVA COMPACT PACKAGED SUBSTATION

All equipment and material shall be designed manufactured and tested in accordance with the latest applicable Indian Standard / IEC standards.

Package Sub-station consisting of 22 KV SF6 Ring Main Unit with breaker as protection, Dry Type Cast Resin Transformer and Low Voltage Switchgear with all connection accessories, fitting & auxiliary equipment in an Enclosure to supply Low-voltage energy from high-voltage system as detailed in this specification.

1. Type of Ventilation: Natural
2. Compartmentalized: Three compartments namely HT Compartment, Transformer Compartment & LT Compartment.
3. Rated temperature enclosure class: K10
4. Degree of protection for the compartment: (a) Transformer Compartment – IP34
   (b) HV & LT Compartment - IP54
5. Applicable Standard: Latest relevant IS / IEC (viz. IS 11171, IEC 62271-200, 61439 1-6, 62271-202 or latest) standards.

The complete unit shall be installed on an OEM recommended plinth (base) as Outdoor substation. The Isolators controls incoming-outgoing feeder cables of the 22 KV distribution systems. The Vacuum Circuit Breaker shall be used to control and isolate the 22 kV/433 V, 250 KVA Distribution transformers. The transformer Low Voltage side shall be connected to Low Voltage switchgear.

(A) ENCLOSURE:

The enclosure shall be made of 2.0 mm thickness Galvanized Sheet Steel tropicalized to meet Indian weather conditions including all the partition sheets & doors. The base of the enclosure shall be of 4.0 mm thickness Hot Dip Galvanized Sheet Steel. The entire Package Substation shall be Factory Assembled & Factory Fitted. The doors shall be lockable type with locking arrangement. The enclosure should take minimum space for the installation including the space required for approaching various doors & equipment inside. The enclosure construction shall be such that it fully protects ingress of rain water, dust & rusting. The enclosure shall be Powder coated with colour specific to Light and Dark Gray combination. There shall be an arrangement for internal lighting operated by associated switch for HV, Transformer & LV compartments separately. In case of combined compact sub stations in which 2 nos. 250 KVA, 22 KV/433V transformers are installed, the enclosure should be designed in such a way that at site 2 compact substations should be coupled with each other and the HT side compartment and LT side compartment should have adequate space to accommodate one HT panel and one combined LT panel respectively and it should be one combined unit.

(B) HT SWITCHGEAR:

Four way SF6 insulated SS enclosure compact Non-extensible RMU, 22 kV 630 Amps, 21 kA for 3 sec. consist of 2 nos. 630 Amps, 21 kA motorized Load Break Switch with integral earth switch having full making capacity shall be used for incoming cables and 2 no. 630 Amps, 21 kA motorized VCB with self powered series trip microprocessor base numerical over current & earth fault relay for protection of 2 nos. 250 KVA transformers & Metering Panel. The RMU should have necessary Mechanical (Ronis Key Interlock) for 2 Incoming supply when operating in Local – Manual mode. Fault passage indicators assembly to be provided IN/OUT side.

Tests - RMU should be internal arc test classification IAC AFLR 21 KA / 1 Sec. All Type / Routine Test According to latest relevant IS/IEC (viz. 60056, 60129, 60265, 60298, 60420, 60529, 60694, 62271 or latest) standards. The contractor shall submit original rest reports from CPRI / ERDA / Any other equivalent Govt. Testing Agencies.

Interconnection: The connection of HT switchgear to Transformer shall be of suitable size of cables and from Transformer to LT switchgear with the help of suitable size and capacity of copper busbars / copper cables as per Railway requirement.

(C) TRANSFORMER:

The distribution transformer shall be of 250 KVA, 22 KV / 433 V, dry type cast resin outdoor type. The transformer shall have ANAN cooling & vector diagram DYN 11 with all equipments and fittings as per IS 11171 or latest.

- TYPE : Dry Type Cast Resin • Ratio : 22 KV / 433 V
- Capacity : 250 KVA
- Vector Group : DYn11
- Off Load Tapping : +5% to -5% in steps of 2.5%
- Winding : Copper
- Class of Insulation : F or above
- Cooling : Air Natural
- Ambient Temperature: - 50°C
- Temperature rise in Winding - 90°C
- No Load Losses - As Per ECBC Norms
- Full Load Losses - As Per ECBC Norms
- Fittings / accessories - Rating & Diagram plate, Base Channel, Earthing terminals, Lifting lugs, Digital winding Temperature Scanner, Indicator etc.
- **Tests** – All Type / Routine Test According to latest relevant IS/IEC (viz. IS 11171/2026, IEC 600726 or latest) standards. In type test Impulse test, HV power frequency withstand test and Temperature rise to be done. The contractor shall submit original rest reports from CPRI / ERDA / Any other equivalent Govt. Testing Agencies.

(D) LT SWITCHGEAR

LT Panel consisting of:

- 02 Nos. Incomer - 400 Amps FP MCCB with Micro Processor Based Release for over load protection, instantaneous short circuit protection, ground fault protection & over temperature protection. Breaking capacity of MCCBs should be 50 KA and Ics = 100% Icu.
- 01 No. 400 Amps Auto transfer switch (with auto, manual and bypass arrangements)
- 04 Nos. Outgoing – 250 Amps FP MCCB with Micro Processor Based Release for over load protection, instantaneous short circuit protection, ground fault protection & over temperature protection. Breaking capacity of MCCBs should be 50 KA and Ics = 100% Icu.
- Digital multi-functioning meter – 02 Nos.
- Metering CTs – 400/5A, Class 1.0, 5 VA - 06 Nos.
- Copper Busbars (Suitable capacity & size) – 1 set
- On/off and RYB indication lamp – 2 sets

Tests - The LT switchgear panel shall be designed manufactured and all type tested in accordance with the latest relevant IEC (viz. IEC 61439 1-6, 60947-2) standards. The contractor shall submit original rest report from CPRI / ERDA / Any other equivalent Govt. Testing Agencies.

(E) EARTHING –

All metallic components shall be earthed to a common earthing point. It shall be terminated by an adequate terminal intended for connection to the earth system of the installation, by way of flexible jumpers/strips & Lug arrangement. The continuity of the earth system shall be ensured taking into account the thermal & mechanical stresses caused by the current it may have to carry. The components to be connected to the earth system shall include:

a) The enclosure of Unitized / prefabricated substation.

b) The enclosure of High voltage switchgear & control gear from the terminal provided for the purpose.
c) The metal screen & the high voltage cable earth conductor.
d) The transformer tank or metal frame of transformer.
e) The frame &/or enclosure of low voltage switchgear
f) Internal Earthing Connection by using 50 X 6 mm G. I. Strips. 63

(F) Tests for PACKAGE Substation:

The Package Substations offered must be all tested as per latest relevant IS/IEC (viz. 62271-202 or latest) standards. The contractor shall submit original rest report from CPRI / ERDA / Any other equivalent Govt. Testing Agencies.

- Routine Tests: a) Voltage tests on auxiliary circuit. b) Functional test. c) Verification of complete wiring.
- Tests to verify the degree of protection.
- Arcing due to internal fault 20 KA/1 Sec. as per latest IEC 62271-202 or latest
- Test to prove enclosure class - Temperature rise of the transformer inside the enclosure.
- Short circuit test to prove the capability of the earthing circuits to be subjected to the rated peak and the rated short time withstand currents.
- Tests to verify the withstand of the enclosure of the prefabricated substation against mechanical stress.
- Internal Arc test for Enclosure.
- Each PSS/CSS completely unit shall be tested to ensure that all of its protective, control and interlock systems are satisfactorily functioning in the manner as required.

2. **For Item No. 3& 4**

Supply, erection, testing and commissioning of 22 KV HT cable of size 3 core 240 sq.mm

Supply, loading, transportation and unloading to site, laying, testing and commissioning of 22 KV grade, 3 core 240 sq.mm XLPE insulated, armoured Aluminum HT cable conforming to relevant IS and specification attached. Price shall also include supply and erection of lugs, clamps, saddles etc. for laying / running, fixing, raising and termination of the cable. The Armour of the cable shall be properly connected to earth with brass glands for proper earthing. Cable shall be laid underground in cable trench, under and across Railway track, along wall / RCC structures etc. as per site requirement. Where ever cables are running parallel both cables shall be laid in the same trench only through separate trunking in such a way that the cable shall not cross each other throughout the length of the trench. Cable identification tags shall be provided throughout the length at every 25 meters interval.

3. **For Item No. 5& 6**

Supply, erection, testing and commissioning of heat shrinkable straight through joints:
supply, loading, transportation, unloading to site erection testing and commissioning of heat shrinkable straight through joint kit suitable for 22 KV, 3 C 240 sq. mm XLPE AL armoured cable, outdoor type as per standard practice & IE Rules with insulated sleeves, additional lugs and nut, bolts for 3 core 240 sq. mm cable.

4. **For Item No. 7 & 8**

Supply, erection, testing and commissioning of heat shrinkable end termination

supply, loading, transportation, unloading to site erection testing and commissioning of heat shrinkable Indoor touch proof termination kit suitable for 22 KV, 3 x 240 sq. mm XLPE AL armoured cable, indoor type as per standard practice & IE Rules with insulated sleeves, additional lugs and nut bolts for 3 core 240 sq. mm cable cables as per site requirement.